authorities successfully raised $2.86 billion in a through a multi-tranche Eurobond issuance in November 2018, which followed a $2.5billion placement in February 2018,

4. The authorities continue to pursue a strategy of moving towards an 40-60 mix between external and domestic debt ratio (currently 30-70). The aim is to lower interest payments and lengthen maturity. However, in line with this strategy to increase the share of external debt, exchange rate risk exposure has been increasing, particularly since Eurobond financing now accounts for 55 percent of external debt financing. Eurobond bullet repayments pose significant refinancing risk when they become due and need to be managed carefully. However, longer maturities associated with Eurobond financing confer significant advantages in minimizing near-term refinancing risk and hence reduce the gross external financing need.

5. Baseline assumptions do not point to a systematic bias in the forecast track record, which is broadly in line with those observed in other surveillance countries (Figure 3). Reflecting the oil price volatility, past projections of the primary balance and growth show some optimistic bias pre-2018.

Stress Tests

6. Debt and gross financing needs do not worsen significantly under the stress scenarios, masking possible unsustainability of liquidity-based indicators. While debt-to-GDP indicators remain sustainable, higher interest rates will further increase Nigeria's vulnerabilities by placing a principal risk on debt service. In particular, an interest rate shock would increase the FG interest-to-FG revenue ratio to almost 125 percent. A standardized combined macro-fiscal shock that captures a combination of shocks, such as to real GDP growth, the deflator, revenues and expenditures, financing costs and exchange rate depreciation, would increase the ratio to above 149 percent. A contingency shock—increasing non-interest expenditures by 10 percent of the banking sector, lower growth by one standard deviation in 2019 and 2020 and increasing inflation and interest cost—would increase the ratio to above 104 percent.

7. The fan charts indicate that there is significant uncertainty around the baseline. The width of the symmetric fan chart, estimated of some 30 percent of GDP, illustrates the degree of uncertainty for equal probability upside and downside shocks (Figure 4). In an asymmetric fan chart, an extreme downside shock that constrains growth to zero results in a more upward-sloping debt path, reflecting a balance of risks skewed to the downside.

Heat Map

8. The heat map identifies market perception as principal risk to debt sustainability, with increased vulnerability compared to the last Article IV assessment.

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Figure 1. Nigeria Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

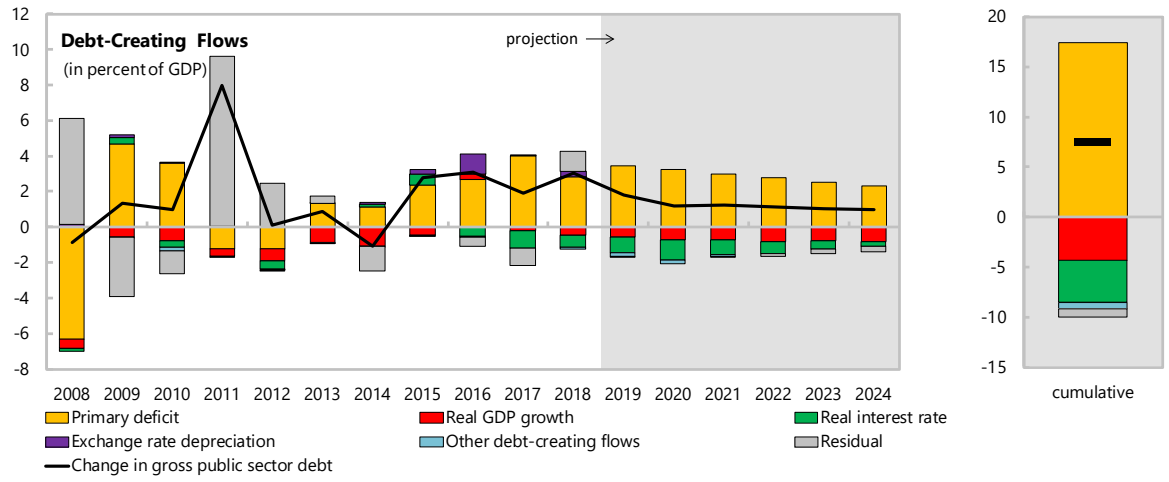
	Actual			Projections					
	2008-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	15.6	25.3	28.4	30.2	31.4	32.7	33.8	34.9	35.9
Public gross financing needs	1.7	5.4	4.7	7.8	7.2	7.1	6.7	6.4	6.7
Real GDP growth (in percent)	5.4	0.8	1.9	2.1	2.5	2.4	2.7	2.6	2.6
Inflation (GDP deflator, in percent)	7.7	11.1	10.2	9.6	8.8	8.4	8.4	8.3	8.1
Nominal GDP growth (in percent)	13.6	12.0	12.4	12.0	11.5	11.1	11.4	11.1	10.9
Effective interest rate (in percent) [4/]	7.8	6.5	7.4	6.4	4.6	5.7	6.4	7.1	7.5
Adjusted Effective interest rate (in percent) [5/]	8.0	7.3	8.4	8.5	6.1	7.9	9.5	11.2	12.5

As of February 22, 2019

Sovereign Spreads		
EMBIG (bp) 3/		496
5Y CDS (bp)		352
Ratings	Foreign	Local
Moody's	B2	B2
S&Ps	B	B
Fitch	B+	B

Contribution to Changes in Public Debt [6/]

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [11/]
Change in gross public sector debt	1.7	1.9	3.1	1.8	1.2	1.3	1.1	1.1	1.0	7.5	
Identified debt-creating flows	0.4	2.9	1.9	1.8	1.2	1.3	1.3	1.3	1.3	8.3	
Primary deficit	0.8	4.1	2.8	3.5	3.3	3.0	2.8	2.5	2.3	17.4	-1.1
Primary (noninterest) revenue and grants	12.1	6.2	8.0	7.0	7.3	7.2	7.3	7.4	7.5	43.6	
Primary (noninterest) expenditure	12.9	10.2	10.8	10.5	10.5	10.1	10.0	10.0	9.8	61.0	
Automatic debt dynamics [7/]	-0.4	-1.1	-0.8	-1.4	-1.9	-1.5	-1.5	-1.2	-1.1	-8.6	
Interest rate/growth differential [8/]	-0.6	-1.2	-1.1	-1.4	-1.9	-1.5	-1.5	-1.2	-1.1	-8.6	
Of which: real interest rate	0.0	-1.0	-0.7	-0.9	-1.2	-0.8	-0.7	-0.4	-0.3	-4.2	
Of which: real GDP growth	-0.5	-0.2	-0.4	-0.5	-0.7	-0.7	-0.8	-0.8	-0.8	-4.3	
Exchange rate depreciation [9/]	0.2	0.0	0.3	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.0	-0.1	-0.2	-0.2	-0.1	0.0	0.0	0.0	-0.6	
Net privatization proceeds (negative)	0.0	0.0	-0.1	-0.2	-0.2	-0.1	0.0	0.0	0.0	-0.6	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroarea	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [10/]	1.3	-1.0	1.1	0.0	0.0	-0.1	-0.2	-0.3	-0.3	-0.8	



Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Defined as interest payments divided by debt stock (excluding guarantees and debt that do not carry interest payments) at the end of previous year.

6/ Gross debt includes overdraft from the central bank.

7/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

8/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

9/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

10/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

11/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

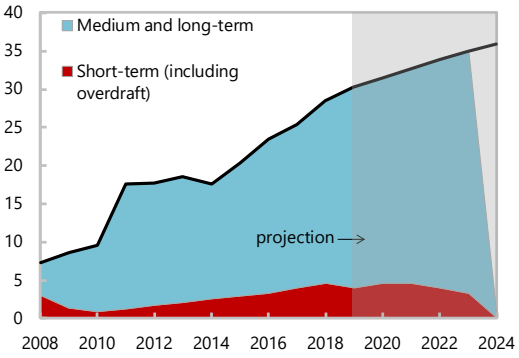
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Figure 2. Nigeria Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

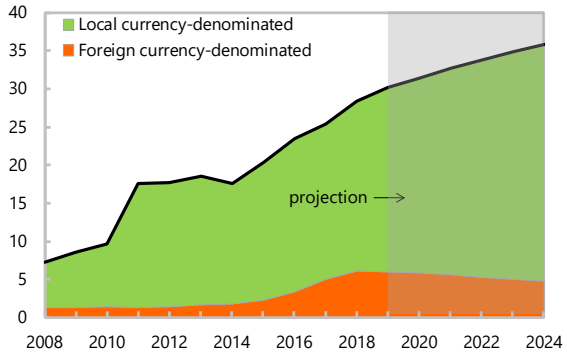
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)

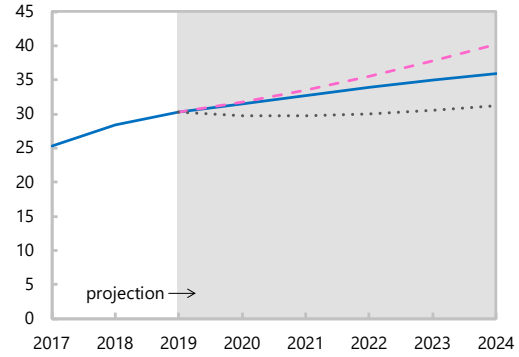


Alternative Scenarios



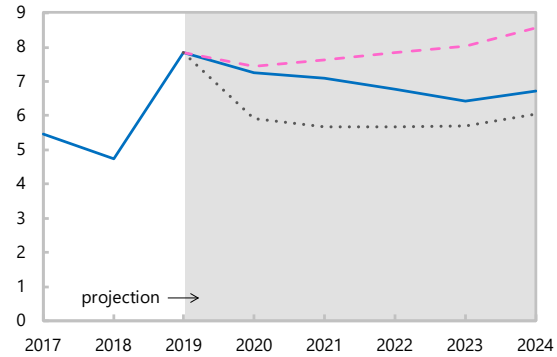
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	2.1	2.5	2.4	2.7	2.6	2.6
Inflation	9.6	8.8	8.4	8.4	8.3	8.1
Primary Balance	-3.5	-3.3	-3.0	-2.8	-2.5	-2.3
Effective interest rate	6.4	4.6	5.7	6.4	7.1	7.5
Constant Primary Balance Scenario						
Real GDP growth	2.1	2.5	2.4	2.7	2.6	2.6
Inflation	9.6	8.8	8.4	8.4	8.3	8.1
Primary Balance	-3.5	-3.5	-3.5	-3.5	-3.5	-3.5
Effective interest rate	6.4	4.6	5.7	6.5	7.3	7.7

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	2.1	4.4	4.4	4.4	4.4	4.4
Inflation	9.6	8.8	8.4	8.4	8.3	8.1
Primary Balance	-3.5	-2.0	-2.0	-2.0	-2.0	-2.0
Effective interest rate	6.4	4.6	5.8	6.6	7.5	8.0

Source: IMF staff.

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